

Core Clinical Trial Documents

1. Clinical Trial Protocol

1.1 Purpose:

The protocol is the master scientific and operational document describing how a clinical trial will be conducted.

It defines:

- Study objectives
- Trial design
- Patient population
- Treatment arms
- Inclusion/exclusion criteria
- Visit schedule
- Safety monitoring
- Statistical methodology

1.2 Connection to Biostatistics

Protocol defines the Primary endpoint, Sample size calculation, Statistical hypothesis, Analysis population

Example:

Primary Endpoint: Progression Free Survival (PFS)

Statistical Method: Kaplan-Meier survival analysis

Analysis Population: Intent-to-Treat

1.3 Key Sections

Section	Purpose
Background	Scientific rationale
Objectives	What question the trial answers
Study Design	Randomization, blinding, control
Endpoints	Measurements used for efficacy
Eligibility Criteria	Who can participate
Statistical Methods	Planned analysis
Safety Assessment	AE/SAE Monitoring

2. Statistical Analysis Plan (SAP)

2.1 Purpose

SAP is the statistical blueprint explaining how collected clinical data will be analyzed.

Usually finalized before database lock.

2.2 Connection to Biostatistics

SAP defines Statistical tests, Missing data handling, Outlier handling, Subgroup analysis, Significance level.

Example:

Endpoint: Overall Survival

Method: Cox proportional hazard model

Effect Measure: Hazard Ratio

Confidence Interval: 95%

2.3 Key Sections

Analysis Populations

Intent-to-Treat (ITT): All randomized subjects.

Purpose: Maintains randomization benefit, Reduces selection bias.

Per Protocol (PP): Only subjects following protocol correctly.

Safety Population: Subjects who received at least one dose.

3. Case Report Form (CRF/eCRF)

3.1 Purpose

CRF is the structured data collection tool used at clinical sites. It converts patient information into analysable clinical datasets.

Example:

Patient Visit: Subject ID, Visit Date, Blood Pressure, Laboratory Results, Adverse Events, Drug Administration



Electronic CRF



EDC Database



SDTM Dataset

3.2 Connection to Clinical Data Management

CDM team performs: CRF design review, Edit checks, Query management, Data cleaning

4. Data Management Plan (DMP)

4.1 Purpose

Defines how clinical trial data will be collected, cleaned, stored, and finalized. It Includes database design, data validation rules, query workflow, coding conventions, data transfer procedures.

Example Workflow:

Site Entry



EDC System



Data Validation



Query Resolution



Database Lock



Statistical Analysis

5. Investigator's Brochure (IB)

5.1 Purpose

Scientific reference document provided to investigators about the investigational product.

5.2 Key Sections

- Preclinical studies

- Pharmacology
- Toxicology
- Previous clinical findings
- Safety information

Prepared by: Sponsor Medical Team

Used by:

- Investigators
- Regulatory authorities
- Ethics committees

6. Informed Consent Form (ICF)

6.1 Purpose

Ensures participant understands: Trial purpose, Risks, Benefits, Alternatives, Rights.

Important principle: No study procedure can happen before consent.

Clinical relevance: ICF protects: Patient autonomy, Ethical conduct, Regulatory compliance

7. Clinical Study Report (CSR)

7.1 Purpose

Final regulatory document summarizing the completed clinical trial.

Submitted during: Drug approval process

Based on: ICH E3 guideline

It contains: Trial design, Patient disposition, Safety results, Efficacy analysis, Statistical outputs

7.2 Structure

Clinical Trial Data



Tables Listings Figures (TLFs)



CSR



Regulatory Submission

8. Adverse Event Documentation

8.1 Purpose

Any unfavourable medical occurrence is termed as Adverse Event (AE).

Example: Headache after treatment

Serious adverse event (SAE) are the emergency life threatening incidences.

Examples: Death, Hospitalization, Life-threatening event, Disability.

Documentation of these events includes: AE Log, SAE Report, Safety Narratives.

8.2 Connection to

Pharmacovigilance + Clinical Data Management

9. Monitoring Plan

9.1 Purpose

Defines how trial quality will be supervised. CRA uses it for: Site visits, Source Data Verification, Protocol compliance checks

It includes:

- Monitoring frequency
- Risk-based monitoring approach
- Escalation process

10. Trial Master File (TMF)

10.1 Purpose

Complete evidence that trial was conducted correctly. It contains as follows:

Zone 1: Trial Management

Examples: Protocol, Monitoring Plan, Meeting Minutes.

Zone 3: Regulatory

Examples: Ethics approval, Regulatory submissions.

Zone 5: Site Management

Examples: Investigator CV, Training records, Delegation logs.

Best Free Learning Resources

Regulatory

- [ICH Guidelines Official Website](#)

Study: ICH E6 GCP, ICH E3 CSR, ICH E9 Statistical Principles

Clinical Data Standards

- [CDISC Official Website](#)

Learn: CDASH, SDTM, ADaM, Define-XML

FDA Resources

- [FDA Clinical Trials Guidance](#)

Study: Electronic records, Clinical trial guidance, Submission standards

Open Clinical Data Practice

- [Vivli Clinical Research Data Sharing Platform](#)
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